Predicting Amazon Household Size from Purchase Histories

UCLA Stats 101C - Regression Project Report

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1 Introduction

Amazon only reliably knows a handful of facts about a customer at sign-up - name, email, phone, age, address - and has no way of knowing how many people actually live in that household. Household size is still useful to know, since it relates to spending capacity and the kinds of products a household needs. This project treats the survey respondents as a stand-in for a third-party data provider that links a household-size label to a person's Amazon order history, and builds a model that predicts household size for the 2,000 Kaggle test respondents whose label is withheld. Since the test set only exposes gender and state, nearly all of the predictive work has to come from purchase behavior itself - order volume, spend, and which product categories a household buys from (toys, baby products, pet food) are the variables I expected to matter most.

2 Exploratory Data Analysis

The response variable, household size, is effectively capped at 4 (the survey question tops out at "4+"), making it a four-level ordinal outcome rather than an open-ended count.

Two-person households are the most common category, with one-person and four-or-more-person households both reasonably well represented and three-person households the least common of the four. This distribution motivated clipping every model's predictions to the 1-4 range, since values outside it cannot occur and would only inflate RMSE.

Larger households place somewhat more orders on average, but the boxes overlap heavily across all four groups, so order count alone does not cleanly separate households by size. This was an early signal that a handful of obvious summary statistics would not be enough, and that more targeted, category-specific features would be needed.

Total spend is heavily right-skewed in raw form, so I applied a log(1+x) transformation before plotting and modeling it; this is one of several numeric-to-transformed-numeric changes used in this project. The log-transformed pattern closely mirrors the order-count pattern above, which is expected given how correlated spend and order volume are with one another, and again does not separate the four groups cleanly on its own.

The share of orders devoted to the toys-and-games category rises steadily with household size and turned out to be the single most correlated numeric feature with the response in this dataset. This matches intuition: larger households are more likely to include children, and toy purchases are a reasonable proxy for that life stage.

This figure illustrates a numeric-to-categorical conversion: I converted the continuous count of baby-category orders into a simple yes/no indicator, since what matters most for this feature seems to be whether a household buys baby products at all, not how many times. Respondents who have made at least one baby-related purchase skew much more heavily toward three- and four-person households than those who have not.

Pet-food share increases mildly with household size as well, though the relationship is noisier than the toy or baby-product signals above. Pet ownership is associated with larger households but is far from exclusive to

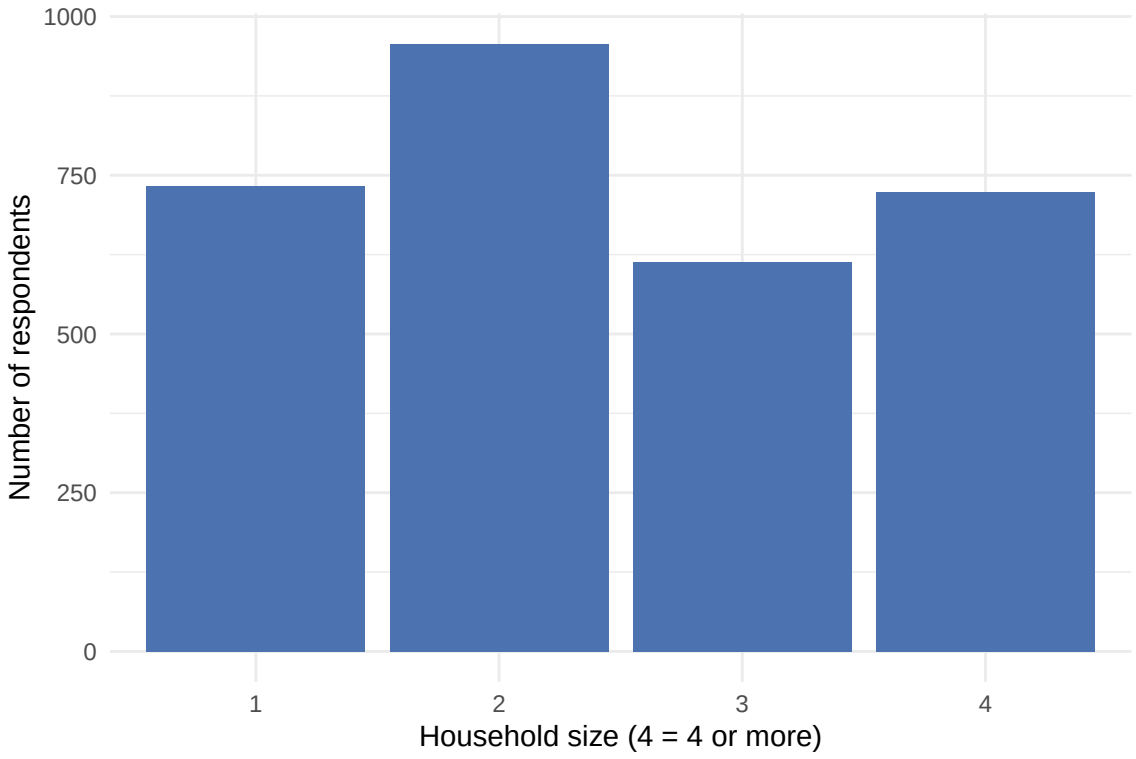


Figure 1: Distribution of household size across the 3,027 labeled training respondents.

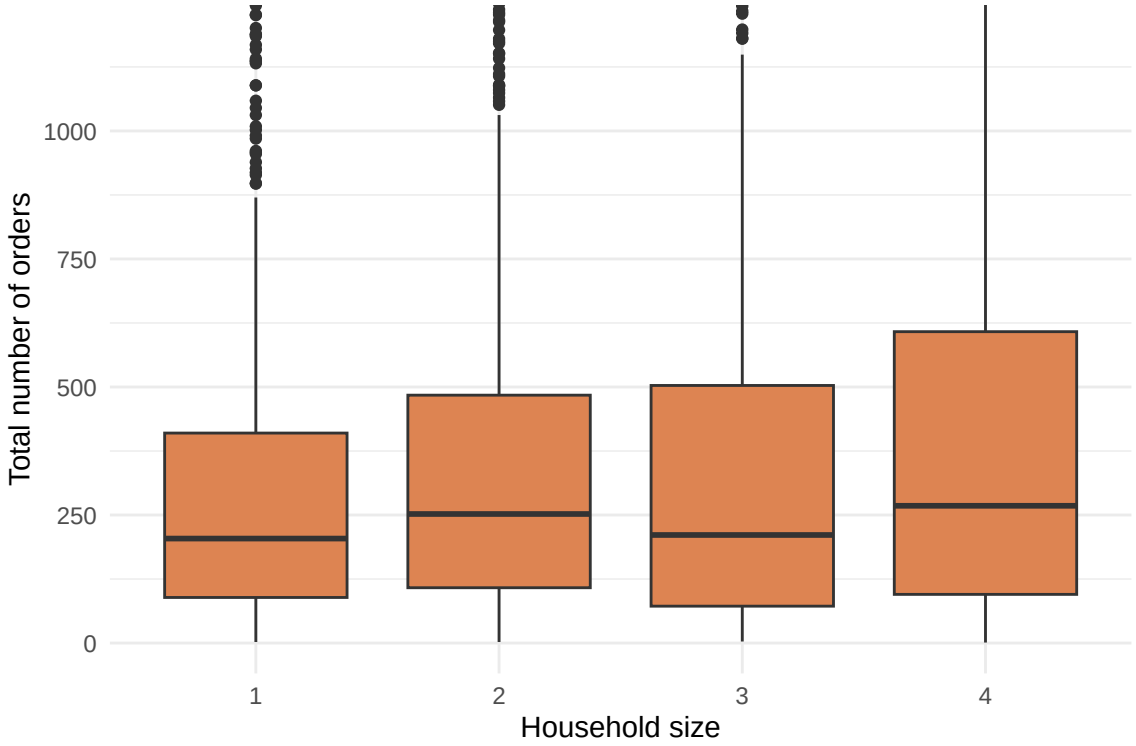


Figure 2: Number of Amazon orders placed, grouped by household size.

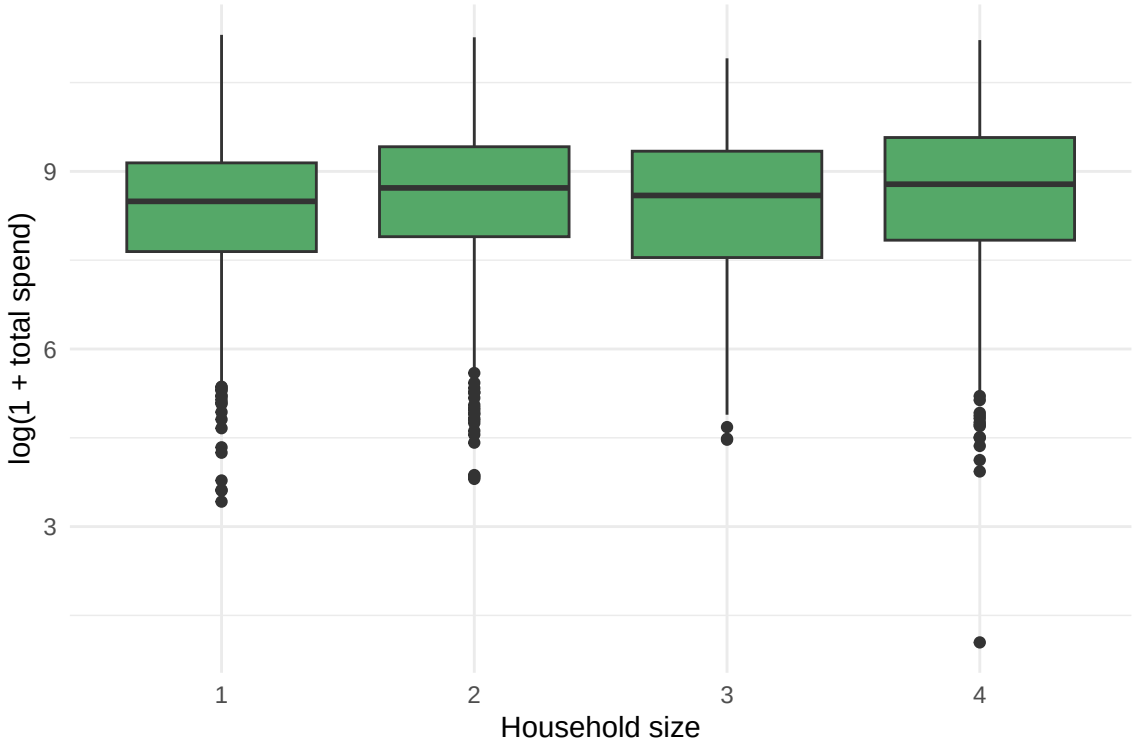


Figure 3: Log-transformed total spend, grouped by household size.

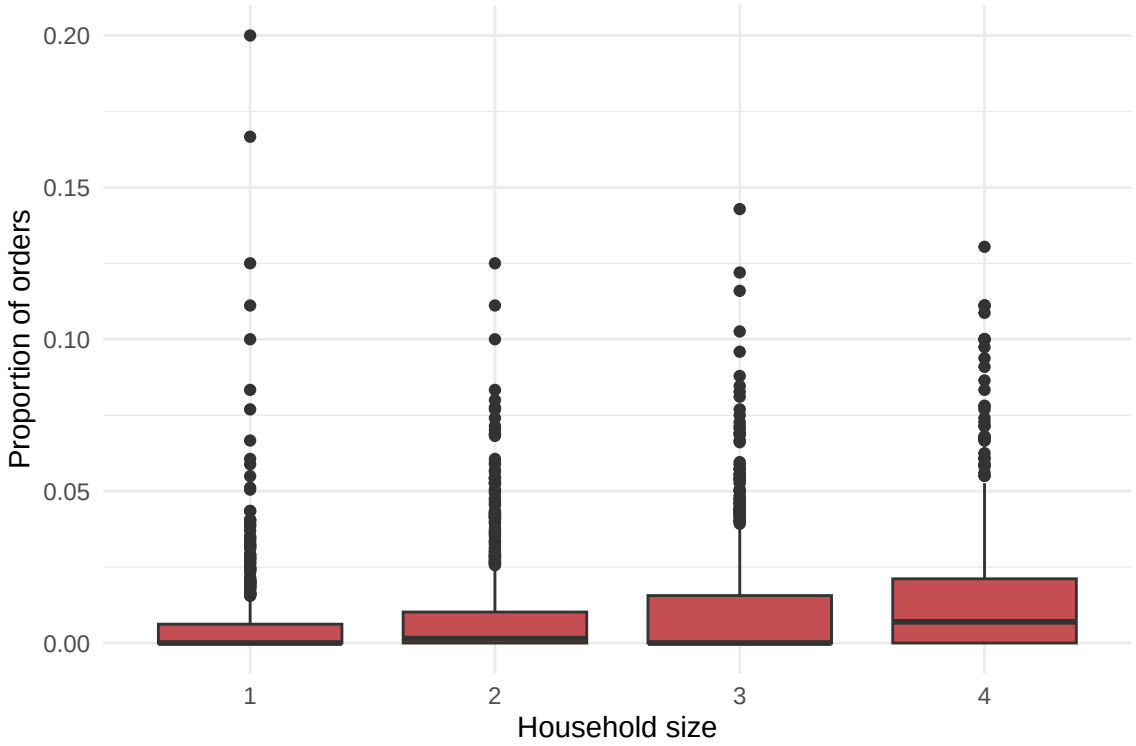


Figure 4: Share of a respondent's orders in the toys-and-games category, by household size.

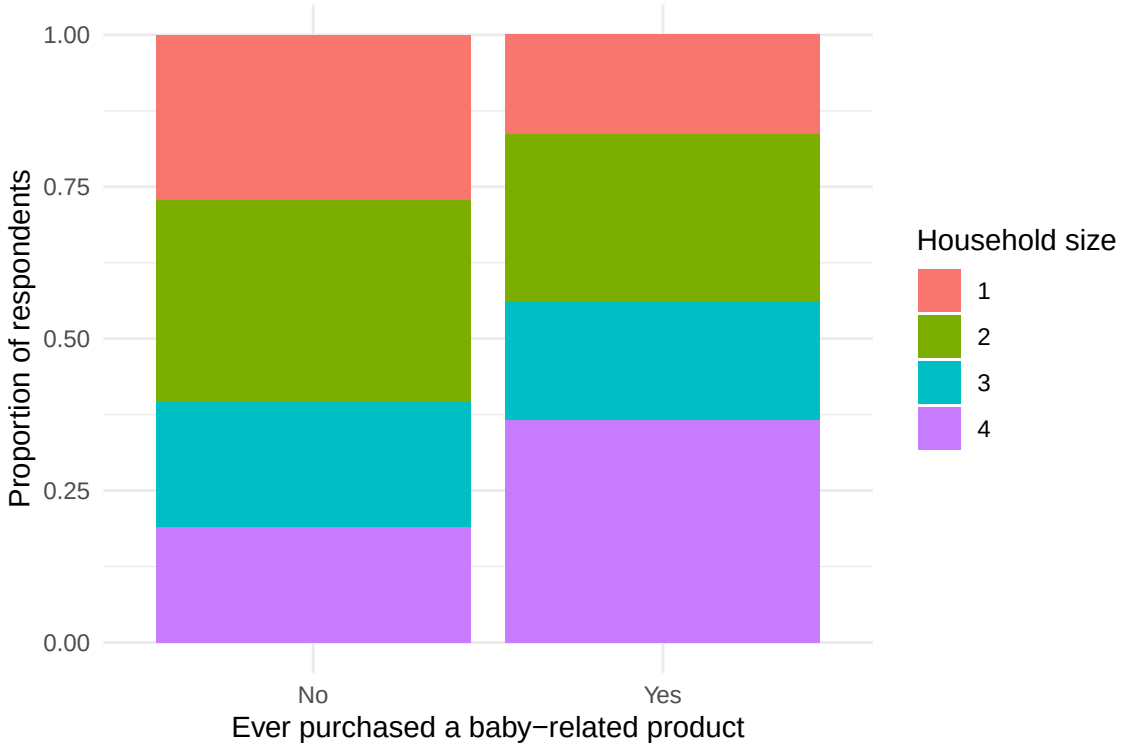


Figure 5: Household-size composition, split by whether a respondent ever purchased a baby-related product.

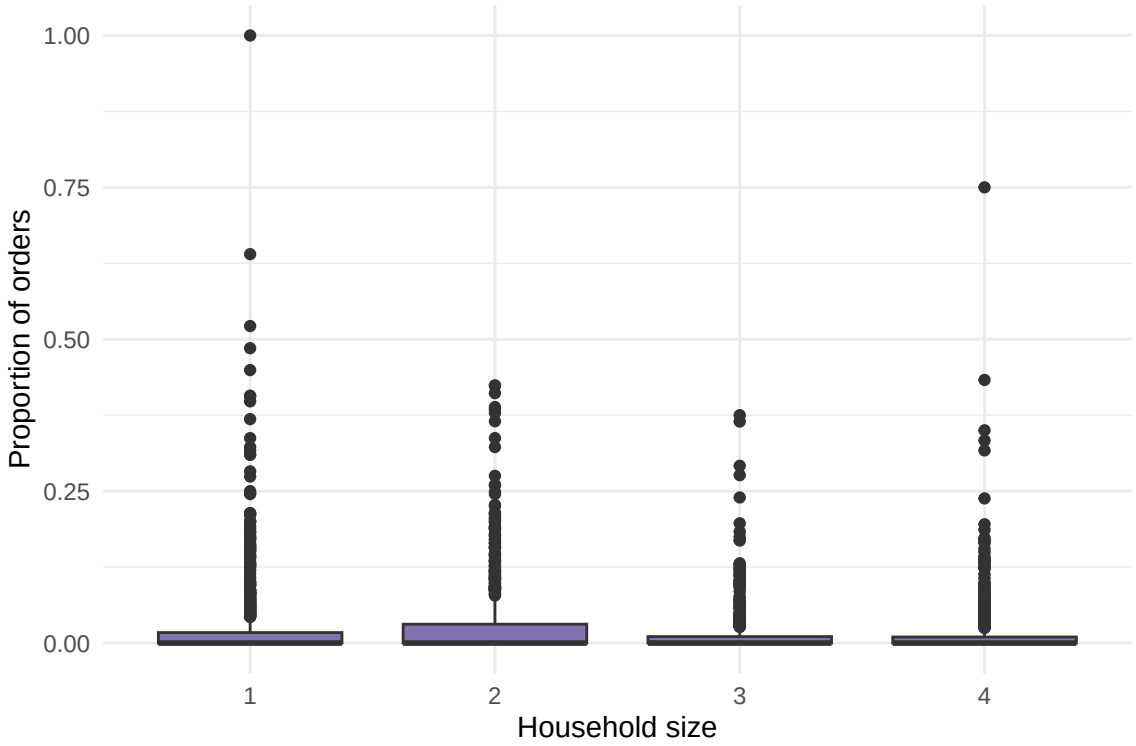


Figure 6: Share of orders in the pet-food category, by household size.

them, so this feature was expected to contribute less to the final model than the two above it.

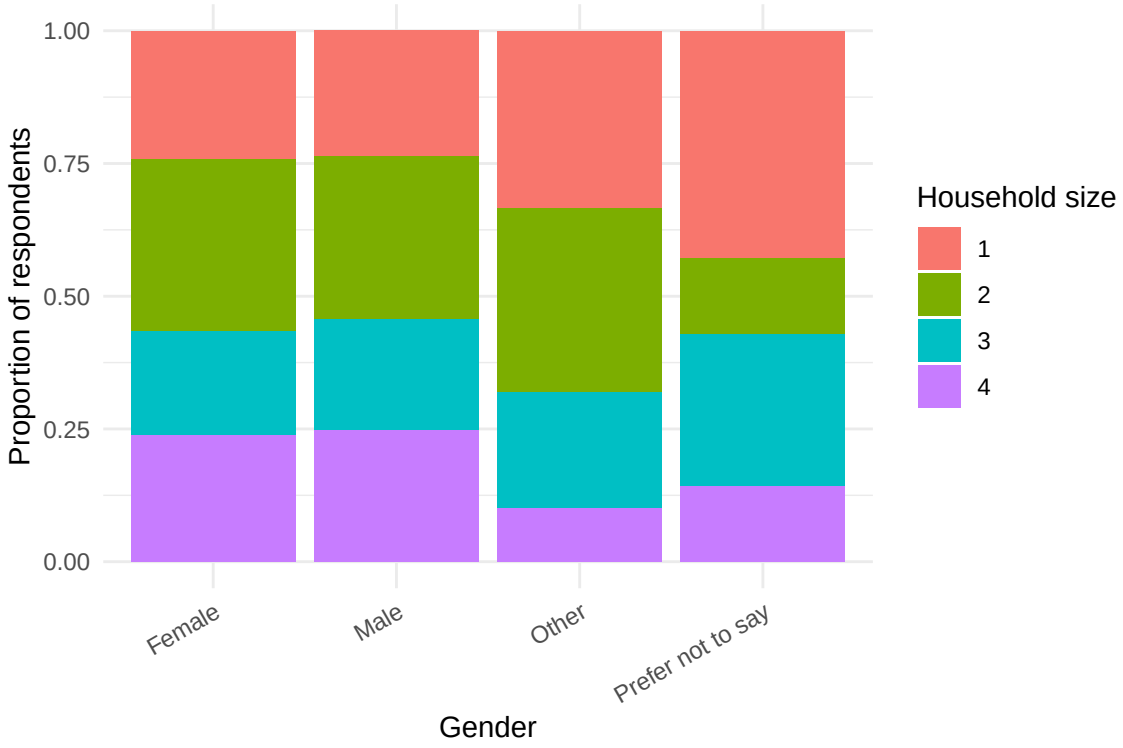


Figure 7: Household-size composition by gender, the only demographic field available in the test set.

Gender is available for every respondent, including the entire test set, but shows only a mild association with household size on its own. It was kept in the model as a low-cost feature rather than a primary driver, since it costs nothing in terms of missing data.

This scatterplot looks at total order count and toy-category share together rather than one at a time, since the two features seemed like they might interact: a high toy share only really signals a larger household when it is backed by a reasonable number of total orders, whereas a single toy purchase out of one or two total orders is much less informative. The respondents shaded for larger household sizes tend to cluster toward the upper-middle region of this plot (moderate-to-high order counts combined with an elevated toy share), which is part of why the tree-based models used later - capable of learning this kind of interaction on their own - outperformed the linear candidates.

Consistent with the boxplots above, category-share features related to toys and family products correlate most strongly with household size among the variables inspected here, while general activity-volume features correlate more weakly. Every correlation shown is under 0.2 in absolute value, which signaled early on that this would be a noisy, high-variance prediction problem best suited to flexible, regularized non-linear models rather than a simple linear fit.

3 Preprocessing

Several preprocessing decisions shaped the final feature set used for modeling. First, respondents with no Amazon purchase history at all (about 10 of the 5,027 total respondents) have missing values for every engineered purchase feature; these were filled with 0 rather than dropped or imputed with a mean, since "zero purchases in this category" is itself a real and informative outcome rather than data that happened to go missing. Second, although `Q.demos.state` is available for every respondent, including the test set,

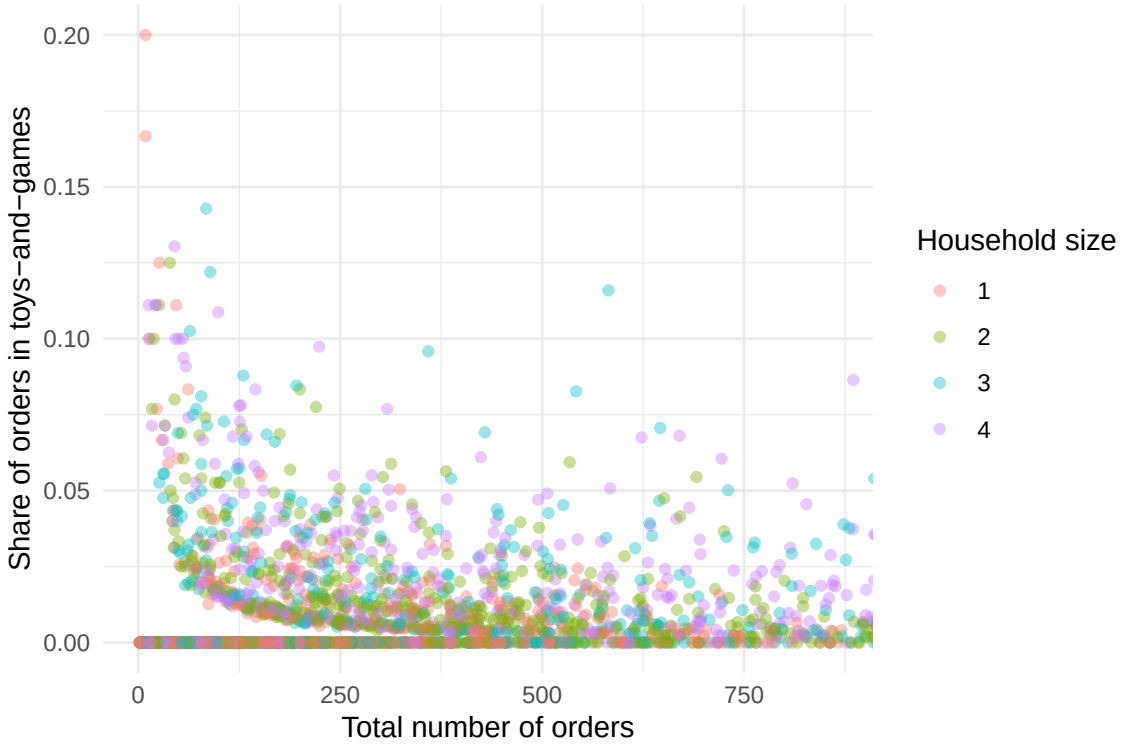


Figure 8: Toy-category order share vs. total order count, colored by household size, illustrating an interaction between the two features.

one-hot encoding it adds roughly 50 mostly-sparse indicator columns. In cross-validation experiments this consistently made RMSE slightly worse rather than better - the extra columns appear to dilute the signal coming from the purchase-based features without contributing meaningful predictive power of their own - so state was excluded from the final feature set, and only gender was retained as a demographic input. Third, and most importantly, after feature engineering the dataset had roughly 180 numeric features against only 3,027 labeled training rows, and none of those features showed a strong individual correlation with the response (see the correlation plot above). This combination - many weak, correlated features and a comparatively small training set - is a classic setup for tree-based models to overfit on noise. To address this, a preliminary random forest was fit on the full training set purely to rank features by importance (using only training labels, with no leakage from the test set), and every candidate model described below was then restricted to the resulting top 45 features. This reduced feature set produced a meaningfully lower cross-validated RMSE than modeling on the full ~180-feature set during experimentation, and is the version carried forward into the candidate model comparison.

4 Candidate Models

Five candidate models were evaluated, all restricted to the same 45 selected features and compared using 5-fold cross-validation on the training set, with predictions clipped to the `[1, 4]` range before scoring to match the possible values of the response.

- **M1_ridge** - a ridge regression, included as a simple, heavily regularized linear baseline.
- **M2_lasso** - a lasso regression, included to see whether an automatic sparse feature selection on top of the already-reduced 45 features would help or hurt.
- **M3_rf** - a random forest, capable of capturing non-linear effects and interactions between features without much manual tuning.

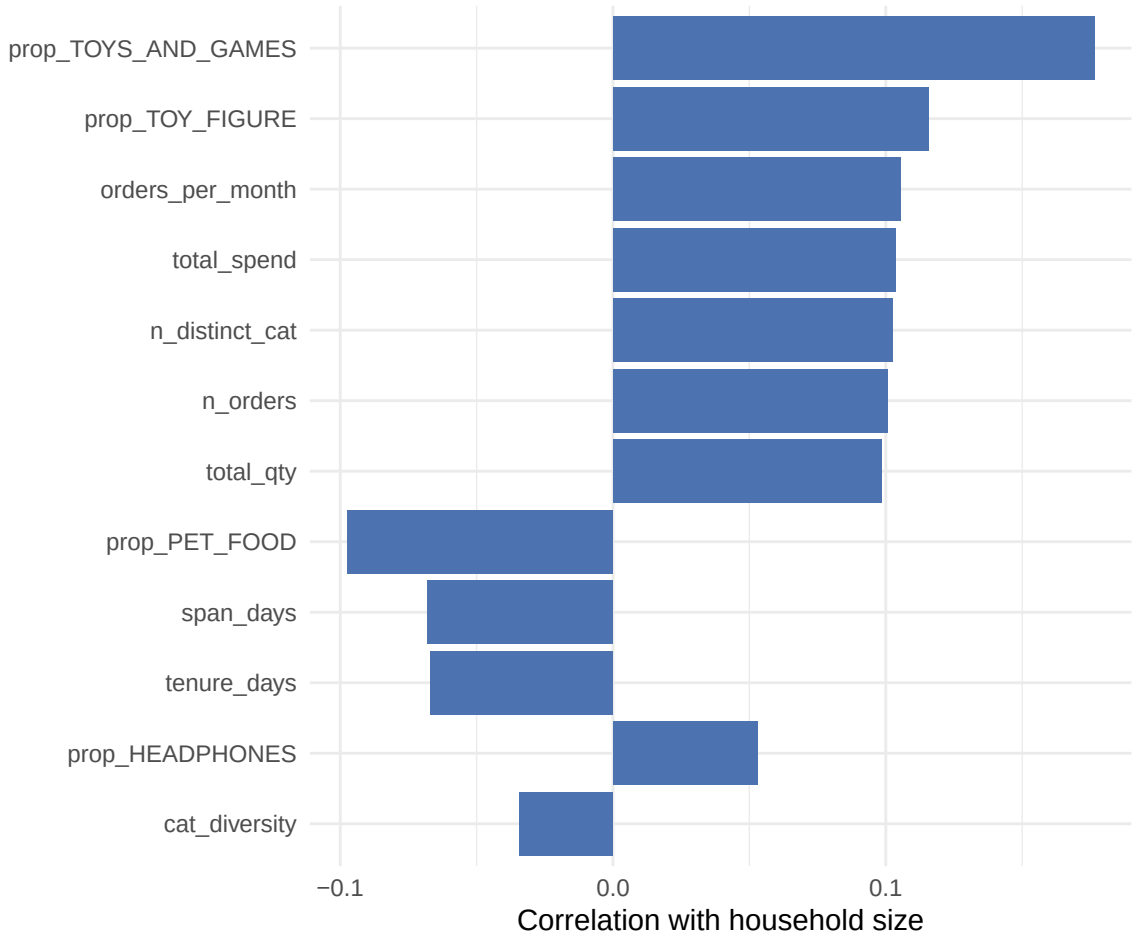


Figure 9: Correlation of selected engineered features with household size.

- **M4_gbm_shallow** - a shallow gradient boosting model, used as a lower-capacity boosting baseline before tuning depth further.
- **M5_gbm_tuned** - a gradient boosting model with a larger tree depth and lower bagging fraction, tuned based on the cross-validation results from the earlier candidates.

Model identifier	Type	Variables	Hyperparameters
M1_ridge	Ridge regression (glmnet)	top 45 features	alpha = 0, lambda chosen by internal CV
M2_lasso	Lasso regression (glmnet)	top 45 features	alpha = 1, lambda chosen by internal CV
M3_rf	Random forest	top 45 features	ntree = 500, mtry = p/3, nodesize = 10
M4_gbm_shallow	Gradient boosted trees	top 45 features	n.trees = 250, depth = 2, shrinkage = 0.02
M5_gbm_tuned	Gradient boosted trees	top 45 features	n.trees = 300, depth = 3, shrinkage = 0.02, bag.fraction = 0.6

5 Model Evaluation and Tuning

	Model identifier	Mean 5-fold CV RMSE
5	M5_gbm_tuned	1.0303
4	M4_gbm_shallow	1.0322
3	M3_rf	1.0324
2	M2_lasso	1.0436
1	M1_ridge	1.0437

Both regularized linear models trail every tree-based model by a noticeable margin, which suggests the relationship between the engineered features and household size is genuinely non-linear and interaction-driven - consistent with the toy-share-versus-order-count scatterplot discussed in the EDA section. `M5_gbm_tuned` achieves the lowest cross-validated RMSE and was selected as the final model; `M4_gbm_shallow` and `M3_rf` are close behind, suggesting the extra tree depth and lower bagging fraction used in the tuned model give it a modest but consistent edge over the shallower boosting model.

Toy- and baby-related purchase shares dominate the top of the importance ranking, alongside general engagement measures such as purchase tenure, order span, and the number of distinct shipping states used. This is consistent with the EDA findings: household-composition proxies carry more weight in the final model than raw order volume alone, and a longer purchase history appears to give the model a more reliable read on a household's typical behavior than a handful of orders would.

Table 3: Three example respondents: values of the top 5 features, actual household size, and the model's prediction.

Respondent ID	Actual	toy orders	TOYS AND GAMES	baby orders	TOY FIGURE	PET FOOD	Predicted
R_zSFZSDB0S1rtmcp	4	0	0	0	0	0.00	2.42
R_3mlroE78njuHGCJ	1	8	0	1	0	0.14	2.02

Respondent ID	Actual	toy orders	TOYS AND GAMES	baby orders	TOY FIGURE	PET FOOD	Predicted
R_e9R9UbMC95GBlp7	1	0	0	0	0	0.00	2.48

For these three respondents, predictions move in the direction implied by their feature values: higher toy- and family-related purchase shares along with longer purchase tenure push the predicted household size upward, while respondents with little activity in those categories receive predictions closer to a one- or two-person household. This individual-level behavior is consistent with the global importance ranking above.

6 Discussion of Final Model

The final model selected is `M5_gbm_tuned`, a gradient boosting model fitted on the 45 features identified as most important by a preliminary random forest screen. It was chosen because it produced the lowest 5-fold cross-validated RMSE among all five candidates, and its advantage over the next-best models held up consistently across folds rather than being driven by a single favorable split.

Its main strength is the ability to capture non-linear, interaction-driven relationships between purchase categories and household size - relationships that the ridge and lasso baselines clearly could not represent, based on their weaker cross-validation scores. It also handles respondents with no purchase history gracefully, since their feature values are simply 0 rather than missing, which the model has effectively learned to treat as a low-activity signal.

Its main weakness is that the underlying signal in this problem is genuinely weak: even the strongest individual feature correlates with household size at well under 0.2, so the final cross-validated RMSE of roughly 1.03 is only a modest improvement over always predicting the training mean, which would produce an RMSE close to the response's standard deviation of about 1.1. Household size is ultimately shaped by factors - family structure, life stage, household income - that purchase history can only partially stand in for, which limits how much further this class of model could realistically improve without additional data sources.

Possible improvements that could plausibly help include ZIP-code-level shipping addresses (to merge in Census household-size statistics for that area), order timestamps at the hour-of-day or day-of-week level (which might expose patterns specific to shared-account usage), and more detailed product metadata such as pack size or explicit "family size" labeling, which the current category-share features can only partially approximate.

7 Appendix: Contributions

This project was completed individually by Guo On Chu, who was solely responsible for the feature engineering, exploratory analysis, candidate model development, cross-validation, hyperparameter tuning, final model selection, and all Kaggle submissions.

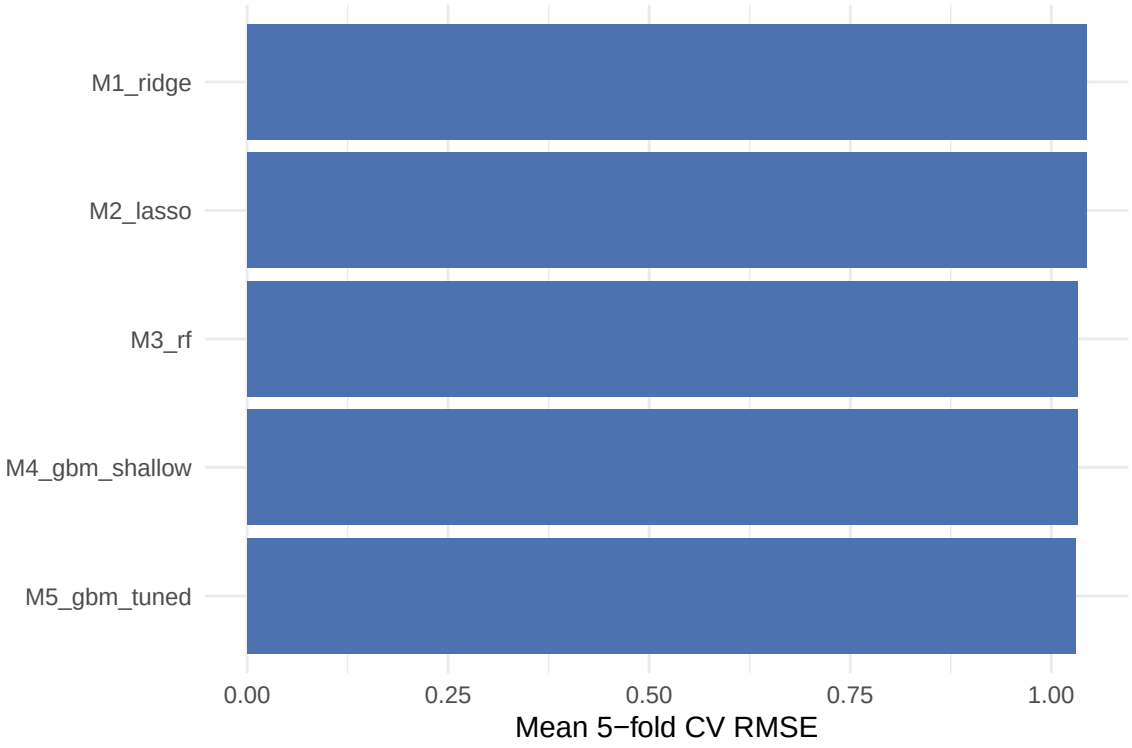


Figure 10: Cross-validated RMSE by candidate model, sorted from best to worst.

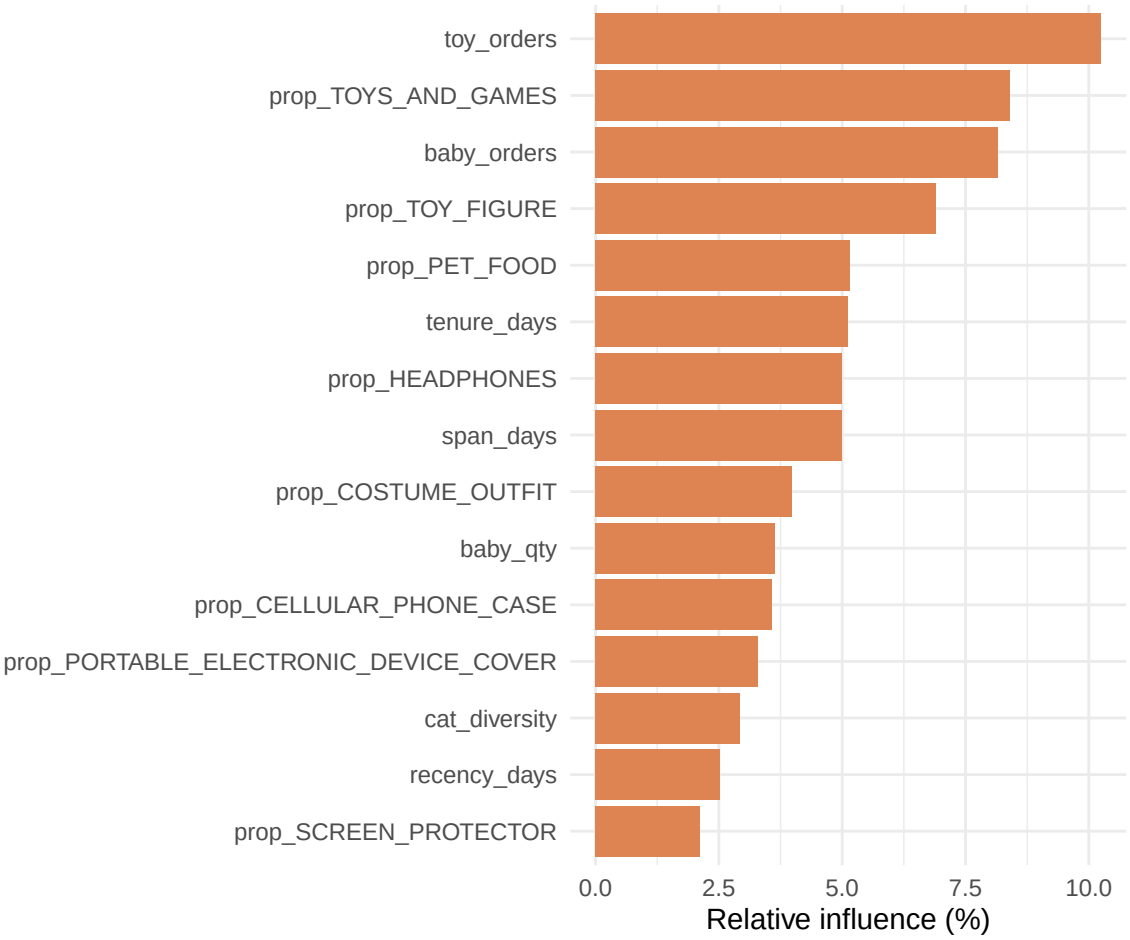


Figure 11: Top 15 feature importances for the selected final model (the tuned GBM), showing the effect of the most important predictor variables.